

Walk $\rightarrow$ Let $G = (V, E)$ be a graph. A finite alternating sequence of vertices and edges, denoted by W , beginning and ending with vertices

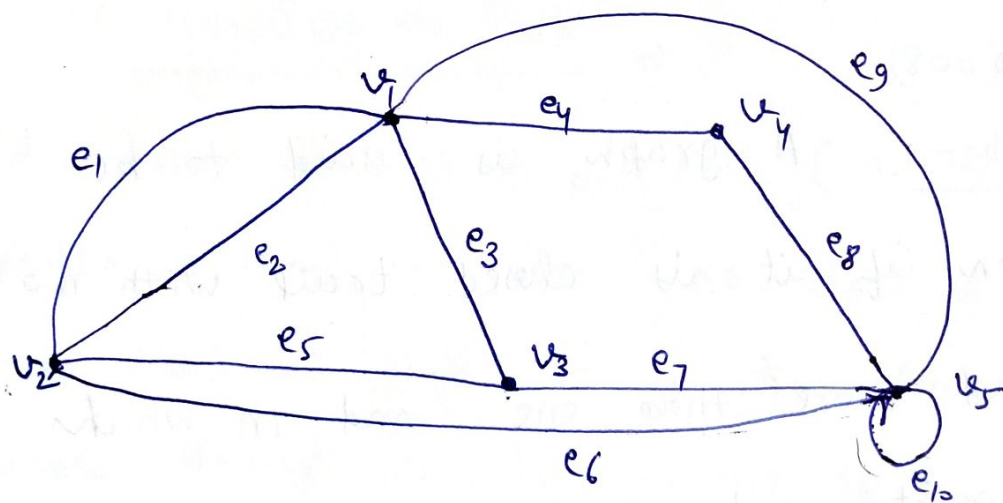
$$W = v_0 e_1 v_1 e_2 v_2 \dots v_{n-1} e_n v_n$$

such that each edge is incident with the vertices preceeding and following it is known as walk $(v_0 - v_n)$

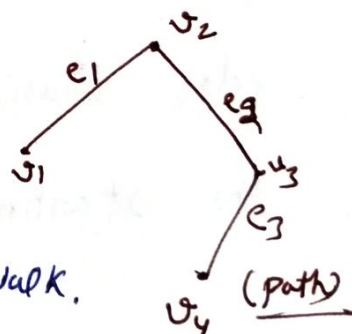
open walk $\rightarrow$ for $v_0 \neq v_n$, the walk $W = v_0 e_1 v_1 e_2 v_2 \dots$

$v_{n-1} e_n v_n$ is called an open walk.

closed walk $\rightarrow$ A walk that began and end with the same vertex is called the closed walk.



$W_1 = v_1 e_4 v_4 e_8 v_5 e_{10} v_5 e_8 v_4 e_4 v_1$ is a closed walk.

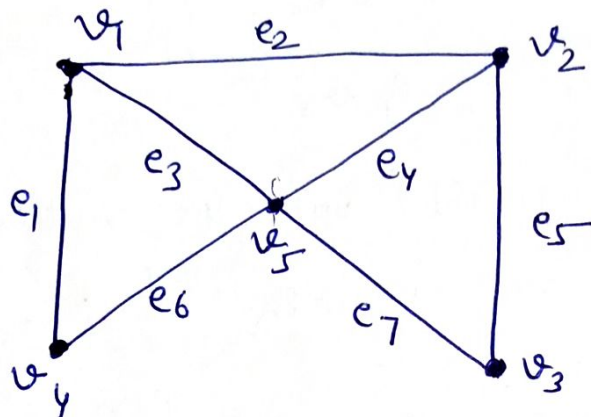


Path $\rightarrow$ An open walk in which no vertex appears more than once is called a path.

$W = v_1 e_3 v_3 e_5 v_2 e_6 v_5 e_8 v_4$ is a path.

Cycles ^(RTU 2012) → A closed path is called a cycle.

In a closed path the end vertices are same.

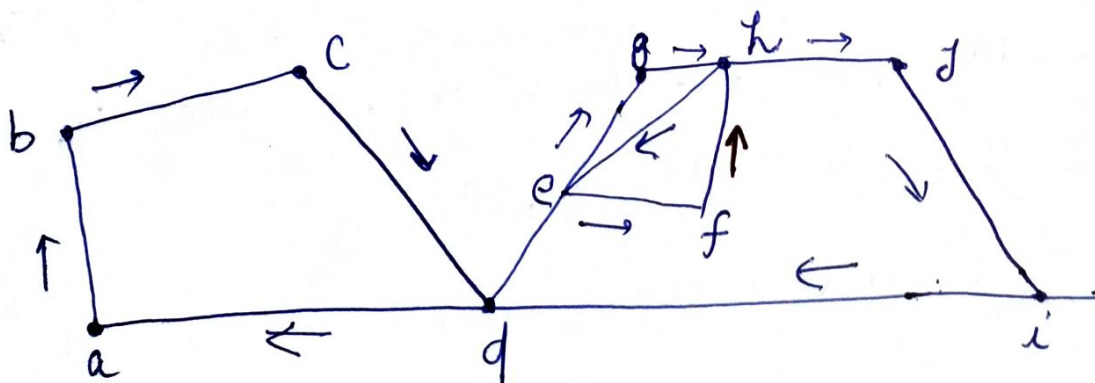


$W = v_1 e_3 v_5 e_7 v_3 e_5 v_2 e_2 v_1$ is a cycle of length 4

$W_1 = v_1 e_1 v_4 e_6 v_5 e_7 v_3 e_5 v_2 e_4 v_5 e_3 v_1$ is a circuit but not a cycle, because here v_5 vertex is repeated.

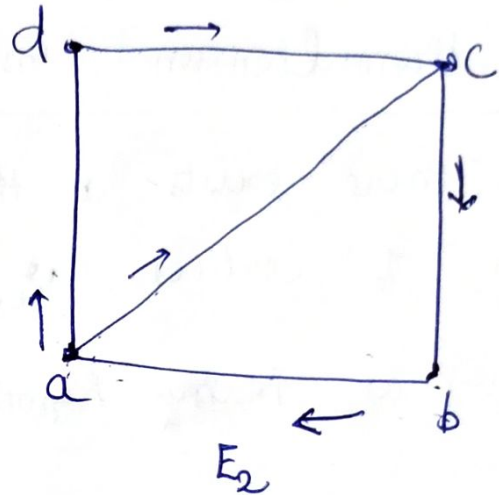
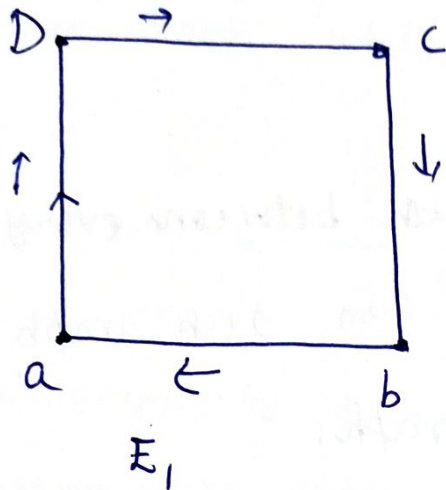
Euler Graph ^{RTU (2008)} →

A graph is said to be Eulerian graph if it is closed trail with no edge transversed more than one and in which the starting point and ending point are common.



* Trail $\rightarrow$ An open walk in which no-edge is repeated, is called a trail. (2)

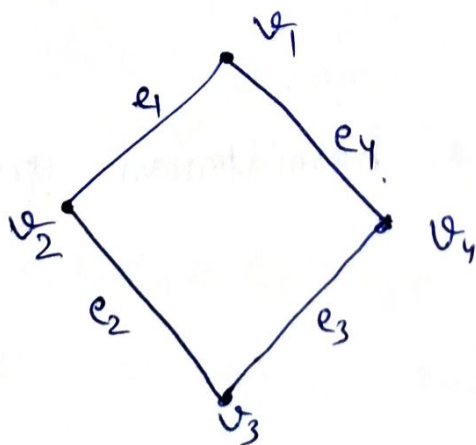
$W_1 = v_1 e_1 v_4 e_8 v_5 e_{10} v_5 e_7 v_3 e_5 v_2$ is a trail



E_1 is a Eulerian graph, while E_2 is not a Eulerian graph

Hamiltonian Path $\rightarrow$

A graph G , a path (a walk in which no vertex is repeated and no edge is traversed more than once) is said to be Hamiltonian path if it passes through all the vertices of G .

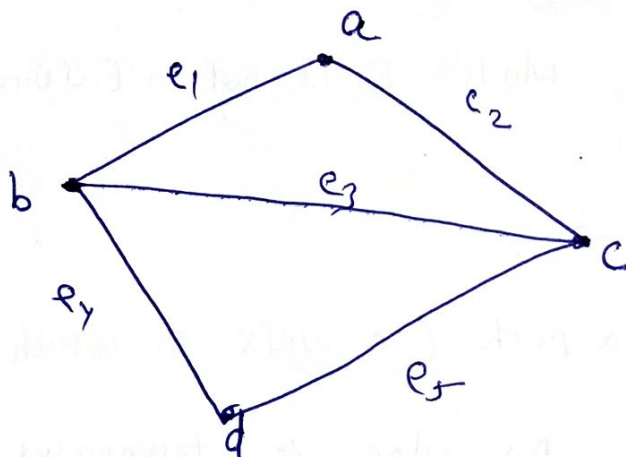


Here $W = v_1 e_1 v_2 e_2 v_3 e_3 v_4 e_4 v_1$, in this walk all the vertices of G , involved no repetition of vertex as well as as edge. So it is closed.

So it is Hamiltonian Path.

Hamiltonian Graph

If there exists a Hamilton path between every pair of vertices $v_i, v_j \in G$, then such graph is known as ~~traverse~~ hamiltonian graph.



$W = a e_1 b e_4 d e_5 c e_2 a$, in this walk all the traversed exactly one. So it is a Hamiltonian path.

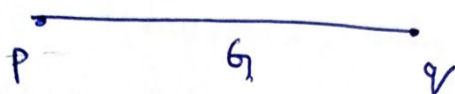
Hence it is a Hamiltonian graph.

(3)

property $\rightarrow$ Let G be a simple graph with n -vertices and let u and v be any two non-adjacent vertices in G , such that

$$\deg(u) + \deg(v) \geq n$$

$\rightarrow$ Draw graph which is neither Eulerian nor Hamiltonian

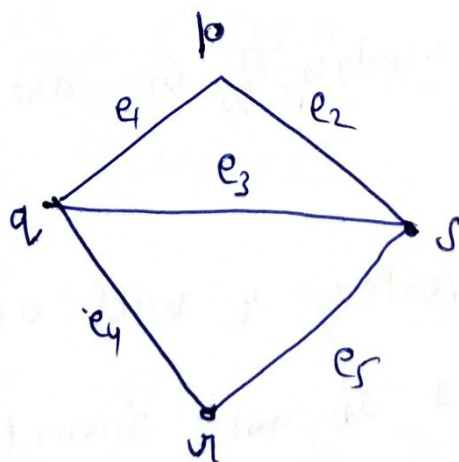


This graph G cannot have any closed walk without repeating the vertices and edges.

Thus neither Eulerian circuit nor Hamiltonian circuit is possible

Hence G is neither Eulerian nor Hamiltonian.

$\rightarrow$ Draw a graph which is Hamiltonian but not Eulerian

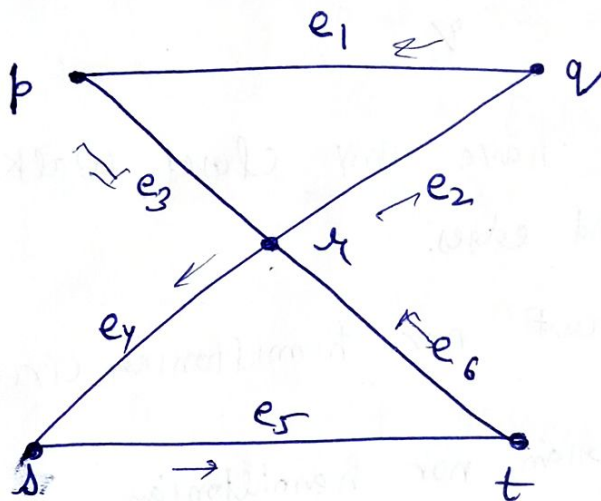


$W = p e_1 q e_4 r e_5 s e_2 p$ be a closed walk. In this walk all the vertices are traversed exactly

once so it is a closed Hamiltonian path. Therefore the graph is Hamiltonian.

On the above walk the edge e_3 is missing, so it is not a closed Euler's line, so the graph is not Euler.

→ Displays a graph which is Euler but not Hamiltonian.

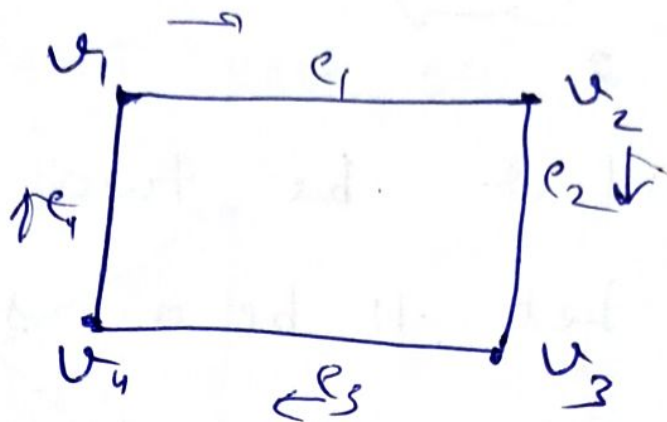


$W = p \xrightarrow{e_3} x \xrightarrow{e_4} s \xrightarrow{e_5} t \xrightarrow{e_6} x \xrightarrow{e_2} q \xrightarrow{e_1} p$ be a closed walk

In this graph all the edges of G are involved
Hence it is Euler.

But in this graph vertex x will occur more than once. ~~Then~~ Hence it is not Hamiltonian graph.

→ Draw a graph which is Eulerian as well as ^④ Hamiltonian



$W = v_1 e_1 v_2 e_2 v_3 e_3 v_4 e_4 v_1$, In this walk each edge is traversed only once.

Hence it is a Eulerian graph.

and In this walk W contains all the vertices and no vertex and no edge is repeated in W .
So it is closed Hamiltonian path.

A simple connected graph (p, q) is Hamiltonian if $q \geq \frac{(p^2 - 3p + 6)}{2}$

Soln Let u and v be two non adjacent vertices of G . Let H be a subgraph of G obtained by removing u and v .

~~vertex~~ club

$$E(H) = q - \text{Deg}(u) - \text{Deg}(v)$$

Maximum no of edges in H

$$= \frac{(p-2)(p-3)}{2}$$

$$\Rightarrow E(H) \leq \frac{(p-2)(p-3)}{2}$$

$$\Rightarrow q - \text{Deg}(u) - \text{Deg}(v) \leq \frac{(p-2)(p-3)}{2}$$

$$\Rightarrow \text{Deg}(u) + \text{Deg}(v) \geq q - \frac{(p-2)(p-3)}{2}$$

$$\Rightarrow \text{Deg}(u) + \text{Deg}(v) \geq \frac{p^2 - 3p + 6}{2} - \frac{(p^2 - 5p + 6)}{2}$$

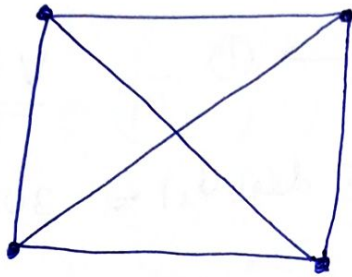
$$\geq \frac{2p}{2}$$

$$\geq p = \underline{\underline{p}}$$

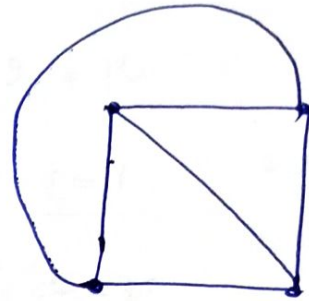
$\Rightarrow$ G is Hamiltonian

Planar Graph → A graph is called planar if it can be drawn in the plane such that no two of its edges intersect each other, except at vertex. (5)

Ex →



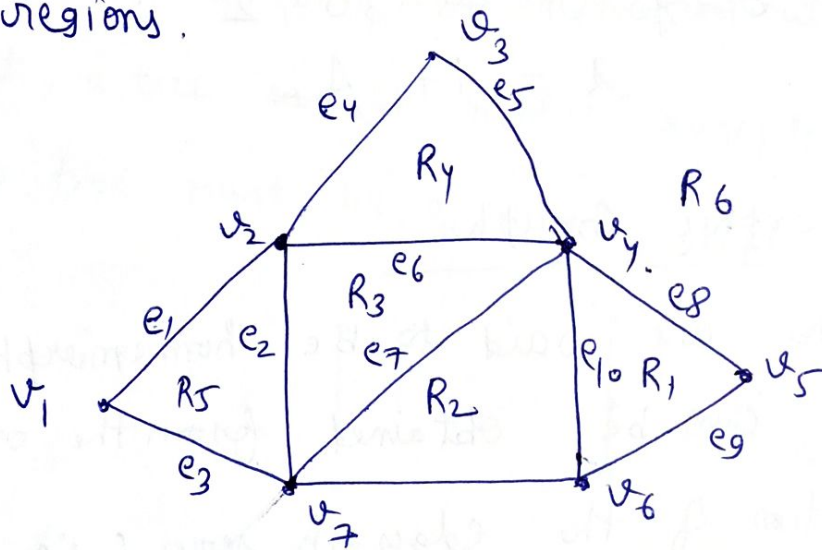
(K_4)



K_4 (with no crossing)

K_4 is a planar graph because it can be drawn without crossings.

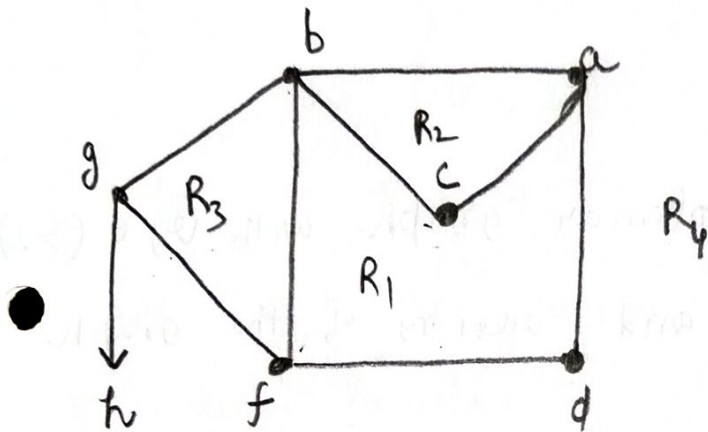
Regions of a graph → A graph divided the plane into regions.



Euler's Formula → Let G be a connected planar graph (Simple graph) with e -edges and v -vertices. If r be the number of regions in a planar graph, then

$$r = e - v + 2$$

Degree of Region $\rightarrow$ Let R be a region in a planar ⁽²⁾ embedding of a graph or a multigraph, the degree of the region R , denoted by $\deg(R)$, is the number of edges transverses in a shortest closed walk about the boundary of R .



In a graph G_1

There are four regions,

$$\deg(R_1) = 5, \quad \deg(R_2) = 3$$

$$\deg(R_3) = 3, \quad \deg(R_4) = 7$$

Since the closed walk ~~is~~ regarding the boundary R_4 is

$$a-b-g-h-f-d-a.$$

Also

$$\sum_{i=1}^4 \deg(R_i) = 18$$

Euler's Formula →

Theorem → Let G be a connected planar simple graph with e edges and v vertices. If r be the number of regions in a planar graph representation of G , then

$$r = e - v + 2$$

(Properties of planar graphs)

Corollary → For a connected planar graph with $v, e (> 2)$ and r as the vertices, edges and regions of the graph respectively, prove that

(i) ~~$\sum \deg(r) = 2e$~~

(i) $\sum \deg(r) = 2e$. (sum of the degree of regions is equal to the twice the number of edges)

(ii) $3r \leq 2e \Rightarrow r \leq \frac{2e}{3}$

(no. of edges regions are less than or equal to $\frac{2}{3}$ of the number of edges)

(iii) $e \leq 3v \Rightarrow$

no of edges are less than or equal to 3 times the no. of vertices

(iv) $e \leq 3v - 6$

(v) $e \leq 2v - 4$

~~Q1)~~ RTU (2009) A planar graph has 30 vertices each of degree 3. (3)

In how many regions can this graph be partitioned?

Soln here $n=3$, and $\sum \deg(v) = 3 \times 30 = 90$

But the sum of the degree of vertices = $2 \times$ no of edges

$$\Rightarrow 90 = 2e$$

$$\Rightarrow e = 45$$

● Let r be the regions of the graphs. then according to Euler's formula,

$$r = e - v + 2$$

$$\Rightarrow r = 45 - 30 + 2$$

$$\Rightarrow r = 17$$

The graph can be partitioned into 17 regions

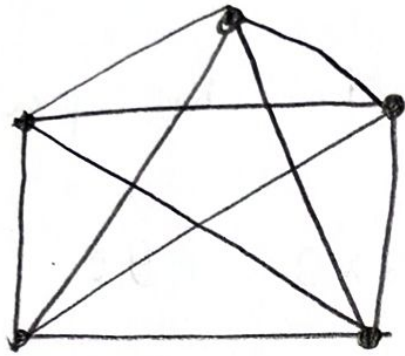
● RTU (2010) (2010)
Kuratowski's Theorem $\rightarrow$

A graph is non-coplanar if and only if it contains a subgraph homeomorphic to $K_{3,3}$ or K_5 .

If G is connected simple planar graph with $n (\geq 3)$ vertices and e -edges and no circuits of length 3, then

$$e \leq 2n - 4$$

Proof →



K₅ Graph

In a graph K_5 as shown in Fig. is the loop free connected graph with 5 vertices and 10 edges.

hence $v = 5, e = 10$.

For a planar graph we know that

$$e \leq 3v - 6$$

$$\Rightarrow 10 \leq 3 \cdot 5 - 6$$

$10 \leq 9$ (which does not satisfies the condition)

that

$$\boxed{e \leq 3v - 6}$$

Thus K_5 is non-planar graph

$$10 \leq 2 \times 5 - 6$$